

Term 3 - COMP0204 Assessment 6 (2025-2025)

Each team is required to submit a GitHub link (via COMP0204 Assessment 6 submission page) with the following details by 29th May. Viva will take place on the 2nd trial run day (2nd June).

The programming component will be assessed through:

- offline review of the GitHub repository and code,
- a short team viva,
- a short live demonstration/test run of relevant robot behaviours.

Required GitHub submission contents

The GitHub repository must include:

- Latest code version (you must also commit the final code version of the finals day - 5th June) - Clearly identify the code used for the viva/test run.
- README file - Explaining the repository structure, required libraries, setup steps, and how to upload/run the code.
- Software overview diagram - Show the main software components and how they interact.
- Flowcharts for key behaviours - Include flowcharts for main behaviours such as line following, RFID/planting, emergency handling, or kill switch.
- Testing/calibration evidence - Include brief notes, logs, screenshots, or videos showing how the code was tested or calibrated. What worked and what did not work.

During the viva: live test run and Q & A

The team may be asked to demonstrate one or more behaviours (followed by a short viva), such as:

- line following,
- RFID reading,
- planting decision,
- motor/actuator control,
- emergency response,
- software kill switch,
- return-to-base logic,
- sensor-based decision-making.

Marking scheme (Total: 20 marks)

1. Github Repository, documentation and algorithm evidence - 5 marks
(marked offline)

5 marks	Repository is accessible and clearly organised. The final version of the code is easy to identify. README explains how to set up, upload and run the code. Repository structure is explained. A meaningful software overview diagram is included. Flowcharts are provided for the main robot behaviours and match the implemented code. Written algorithm explanation is concise and useful.
3-4 marks	Repository is accessible and contains the final code. README and structure are mostly clear, but some setup details may be missing. Diagrams or flowcharts are included but may be incomplete, slightly unclear, or not fully connected to the code.
1-2 marks	Repository exists but is difficult to navigate. README is minimal. Final code may be unclear. Diagrams or flowcharts are missing, superficial, or confusing.
0 mark	No usable repository, inaccessible link, missing code, or no meaningful documentation.

2. Code quality, readability and maintainability - 5 marks
(marked offline)

5 marks	Code is well structured, readable and maintainable. Major behaviours are separated into functions or files. The main loop is easy to follow. Names are meaningful. Constants are used for thresholds, timings and limits. Comments explain non-obvious logic.
3-4 marks	Code is mostly understandable. Some modularity is present. There may be long functions, duplicated logic, inconsistent naming, sparse comments, or some hard-coded values.
1-2 marks	Code may work but is difficult to follow. Most logic is inside one large loop(). Many unexplained numbers are used. Comments are limited or unhelpful.
0 mark	Code is missing/unreadable/cannot be meaningfully assessed.

3. Live testing, calibration and viva response - 4 marks
(marked during the viva)

4 marks	Team provides clear evidence of testing and calibration. They can explain the code confidently and connect code decisions to robot behaviour. They honestly discuss
---------	---

	limitations and realistic improvements.
2-3 marks	Some testing and calibration evidence is shown. Team can explain most code decisions, but some details are vague or only one member can answer.
1 mark	Limited testing evidence. Team gives superficial explanations. Weak understanding of why the code works or fails.
0 mark	No testing evidence and team cannot explain the code.

4. Sensor, actuator and mission functionality - 3 marks **(marked during the viva)**

3 marks	Code reads and interprets relevant sensors correctly. Motor/actuator outputs are controlled reliably. Mission behaviours such as line following, RFID reading, planting decision, return behaviour, or revival-related outputs are implemented.
2 marks	Most key inputs and outputs are implemented, but some behaviours are incomplete, unreliable, or not fully integrated.
1 mark	Basic motor control or one sensor behaviour exists, but integration is weak. The robot may rely heavily on assumptions or manual positioning.
0 mark	No meaningful sensor/actuator integration is demonstrated.

5. Autonomy, control logic and safety handling - 3 marks **(marked during the viva)**

3 marks	The robot has clear autonomous logic. Behaviour transitions are sensible and triggered by events, sensor readings or server messages. Emergency and kill-switch logic have high priority.
2 marks	Autonomous structure is present but incomplete. Some transitions work, but others are fragile or based on timing/manual assumptions. Safety handling exists but may not be checked frequently enough.
1 mark	Robot behaviour is mostly scripted. There is limited response to events. Emergency or kill-switch behaviour is minimal or unreliable.
0 mark	No meaningful autonomous logic or safety handling.